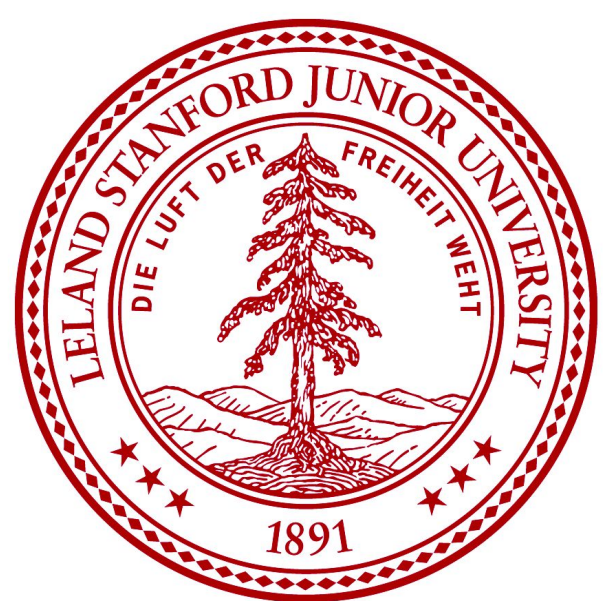
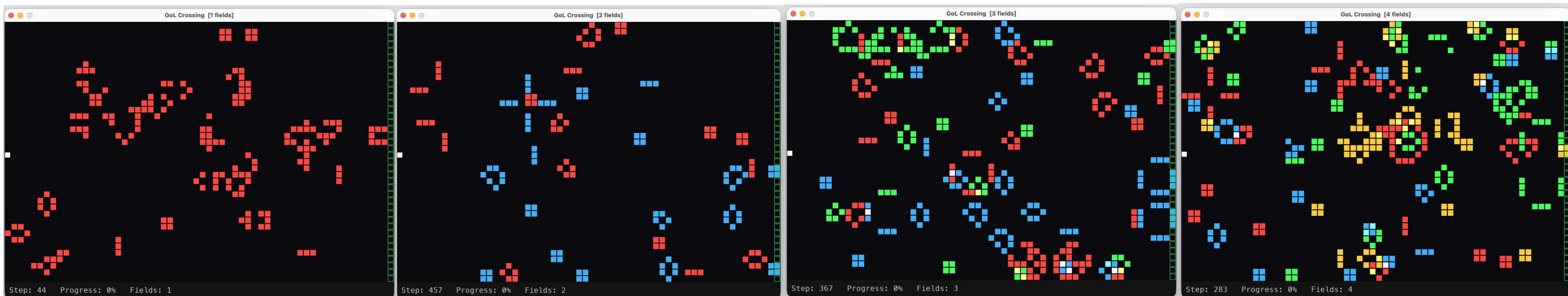




Improving Embodied Navigation with Next-Frame Prediction on Synthetic Video Sequences

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Overview

Video pretraining captures environment dynamics. Next-frame prediction allows a model to learn how the world evolves, potentially internalizing transition rules.

Video pretraining transfers to embodied control. Models trained on video can acquire priors that improve robot navigation and manipulation, suggesting next-frame prediction can capture actionable knowledge about the world.

Need for controlled evaluation. Real-world benchmarks conflate visual complexity with dynamics understanding. A synthetic setting with known ground-truth transition rules enables precise, unambiguous measurement of what the model has actually learned.

Can video pretraining serve as a foundation for embodied navigation?

Setup

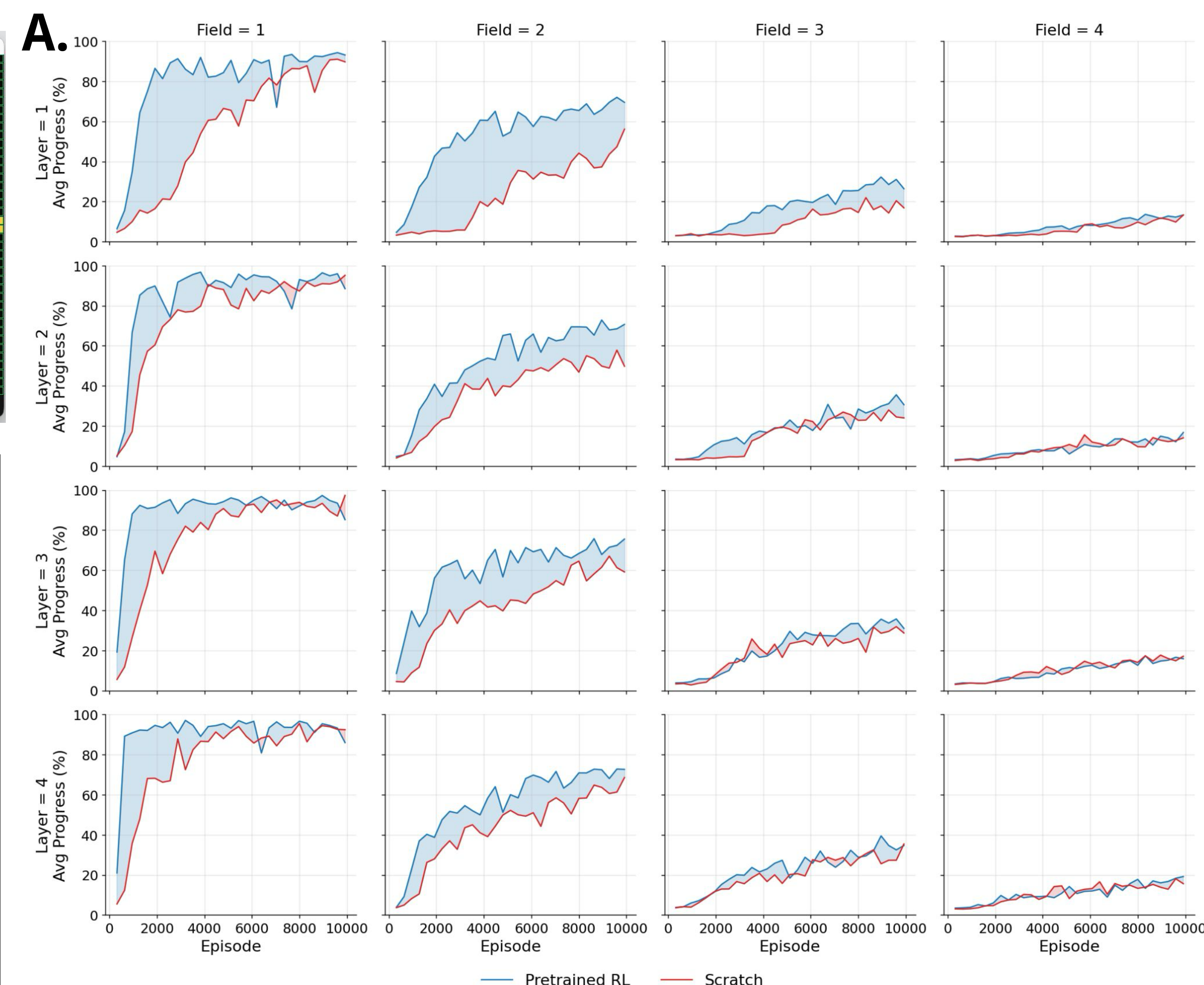
Testbed: Game of Life grid worlds. We use multi-field Game of Life variants, where an agent must understand spatial structure and anticipate temporal transitions to navigate safely from one side to the other.

Multi-field Game of Life as synthetic video. We run K parallel Game of Life fields ($K = 1\text{--}4$) on a 96×96 grid, each color-coded in the RGB display.

Frame Crossing Game (RL task). The agent starts on the left edge and must reach the right side. Touching any active Game of Life cell ends the episode.

Two-stage pipeline. (1) Self-supervised pre-training via next-frame prediction on synthetic Game of Life videos, then (2) reinforcement learning in the frame-crossing game environment.

Our model is a spatial-conv Transformer that applies 3×3 depthwise convolutions per frame followed by multi-head self-attention over time, predicting the next frame as a per-cell classification task.



Results

- A.** The difference in episode progress for the pretrained runs versus the runs trained from scratch. Pretraining runs outperform those trained from scratch.
- B.** The change in performance as the task difficulty and the model size increase. Increasing with model size and decreasing with level of difficulty as expected.
- C.** Validation loss for pre-training runs. Each subplot demonstrates a different level of difficulty, while the colors show model size.
- D.** Comparison between the two reward functions. No major performance differences between the two rewards.

